

Unit 20

POLYMERISATION

27 items · 65 marks

Syllabus sections in this unit

20.1 Addition polymerisation

Unit 20: Polymerisation

1. M/J 26/P21/Q5/f

- 5 Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

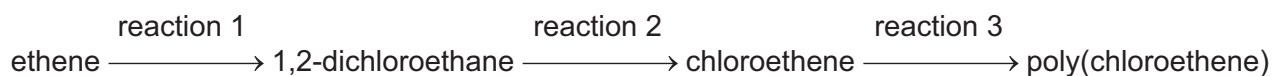


Figure 5.1

- (f) Draw the repeat unit of poly(chloroethene).

[1]

2. M/J 26/P24/Q5/b(i,ii)

- 5 Long-chain alkanes from crude oil can be cracked to make alkenes.

(b) Propene is an alkene that is made by cracking. Alkenes are used in addition polymerisation.

- (i) Draw **one** repeat unit of the addition polymer formed from propene.

[1]

- (ii) The disposal of poly(alkenes) by combustion produces harmful products.

State **one** other disadvantage associated with the disposal of poly(alkenes) such as poly(propene).

[1]

3. O/N 25/P21/Q4/b(i,iii,iv)

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

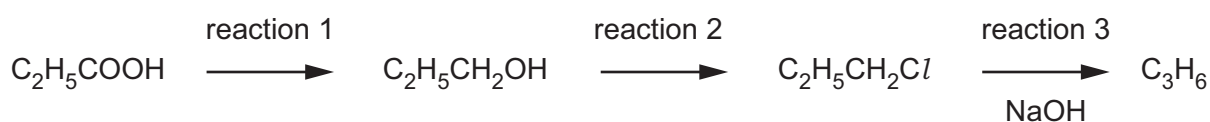


Fig. 4.1

(b) Under suitable conditions, propene polymerises to form poly(propene).

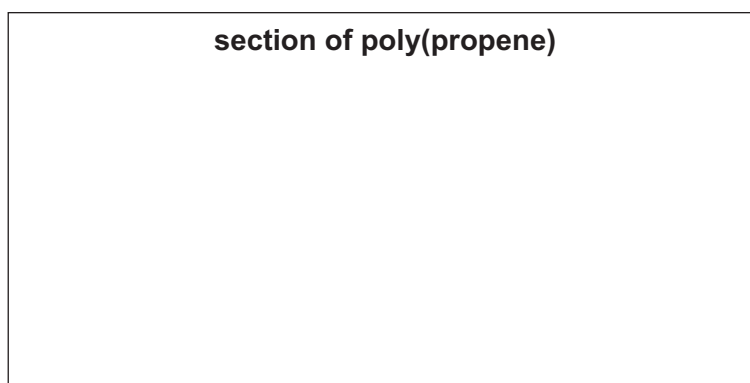
Poly(propene) exhibits stereoisomerism.

(i) Identify the type of polymerisation that forms poly(propene) from propene.

[1]

(iii) Draw a section of poly(propene), showing **two** repeat units.

Use your diagram to identify the type of stereoisomerism shown by poly(propene).
Explain your answer.



[3]

(iv) State **two** difficulties associated with the disposal of poly(propene).

1

2

[2]

4. O/N 25/P23/Q4/b(i,iii,iv)

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

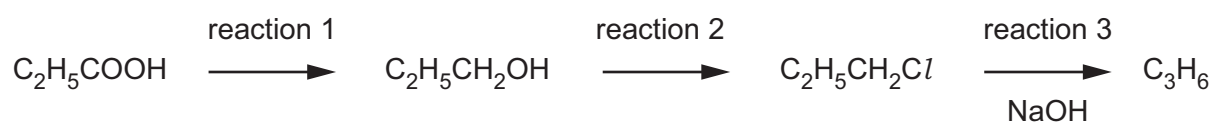


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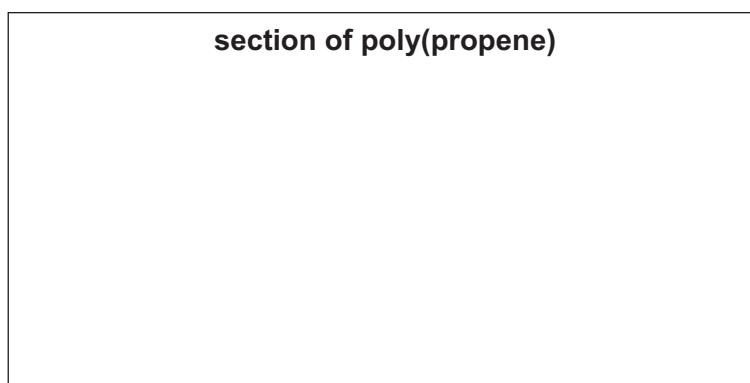
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[1]

(iii) Draw a section of poly(propene), showing **two** repeat units.

Use your diagram to identify the type of stereoisomerism shown by poly(propene).
Explain your answer.



[3]

(iv) State **two** difficulties associated with the disposal of poly(propene).

1

2

[2]

5. M/J 25/P23/Q5/b(vi),c

5 (a) **W** is a colourless liquid.

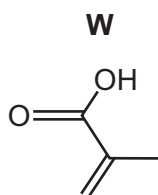


Fig. 5.1

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.

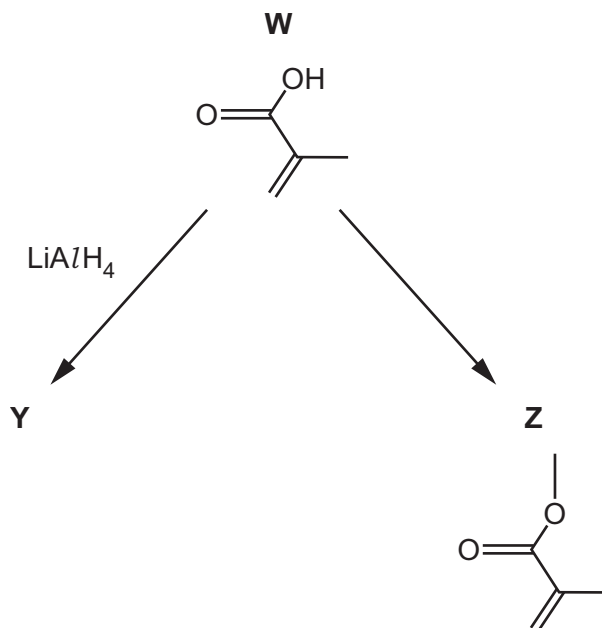


Fig. 5.2

(vi) **Z** is used to produce addition polymer **Q**.

Draw the repeat unit of polymer **Q**.

[1]

(c) Poly(ethene) is an addition polymer made from ethene.

Explain why ethene reacts with electrophiles but poly(ethene) does **not**.

[1]

6. MAR 24/P22/Q2/d(iii)

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

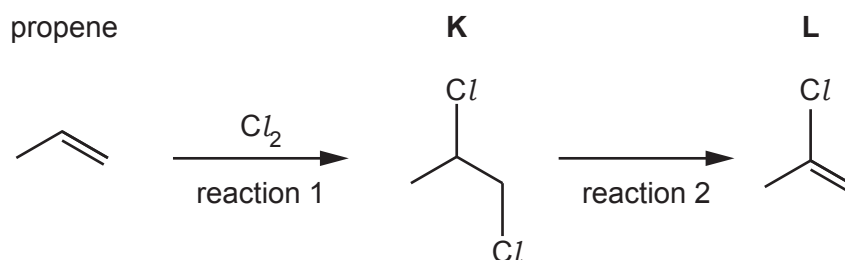
Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

**Fig. 2.1**

(iii) Draw **one** repeat unit of the addition polymer that forms from **L**.

[1]

7. O/N 23/P21/Q4/a(v)

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10} .

**Fig. 4.1**

(v) **C** can form an addition polymer.

Draw the structure of **one** repeat unit of this addition polymer.

[1]

8. O/N 23/P23/Q4/a(v)

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10} .

**Fig. 4.1**

(v) **C** can form an addition polymer.

Draw the structure of **one** repeat unit of this addition polymer.

[1]

9. M/J 23/P23/Q6/a

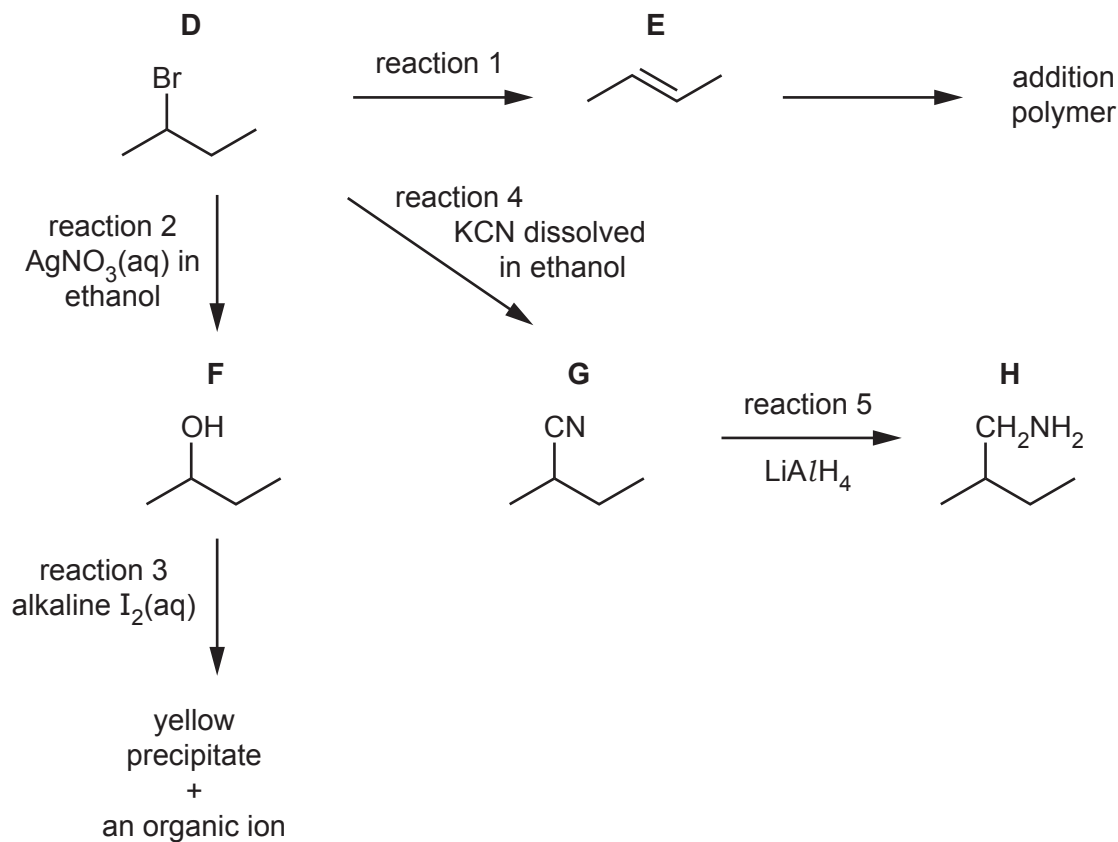
6 Compound **W**, $CH_2=CHCN$, is used to make an addition polymer which is present in carbon fibres.

(a) Draw **one** repeat unit of the addition polymer of **W**.

[1]

10. MAR 23/P22/Q4/a(ii)

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

**Fig. 4.1**

(ii) Draw the structure of **one** repeat unit of the addition polymer that forms from **E**.

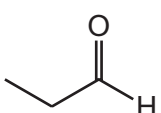
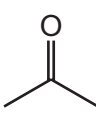
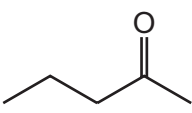
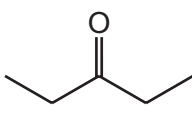
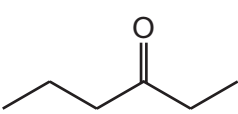
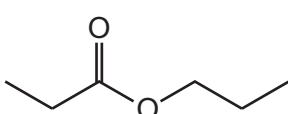
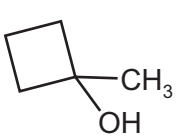
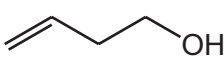
[1]

11. O/N 22/P22/Q3/b

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

(a) Table 3.1 shows four pairs of organic compounds.

Table 3.1

organic compounds		reagent	positive result of chemical test on identified compound
A1 	A2 		
B1 	B2 		
C1 	C2 		
D1 	D2 		

(b) **D2** forms polymer **Z** when heated gently.

(i) Identify the type of polymer that forms from **D2**.

[1]

(ii) Draw one repeat unit of polymer **Z**.

[2]

12. M/J 22/P22/Q4/c

(c) **X** is an addition polymer.

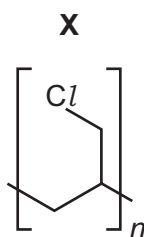


Fig. 4.3

(i) Draw the monomer of **X**.

[1]

(ii) Suggest **one** reason why the disposal of items made from **X** is difficult.

[1]

13. MAR 22/P22/Q4/b(ii)

4 Compounds **J** and **K** are found in plant oils.

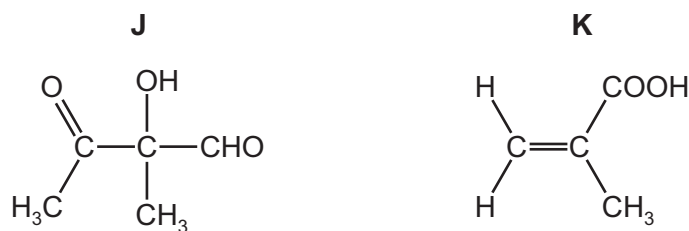


Fig. 4.1

(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.

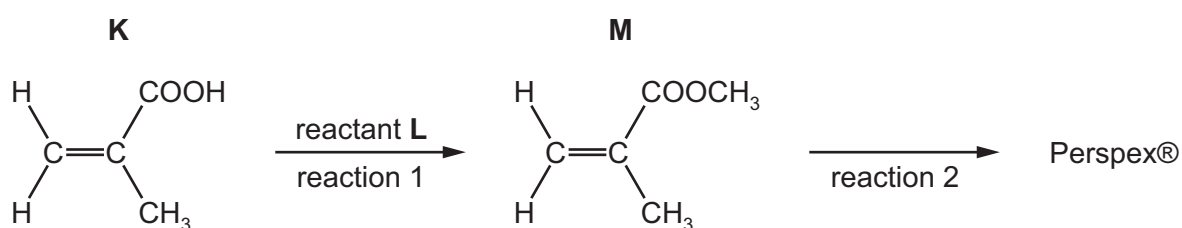


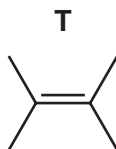
Fig. 4.2

(ii) Draw **one** repeat unit of the addition polymer Perspex®.

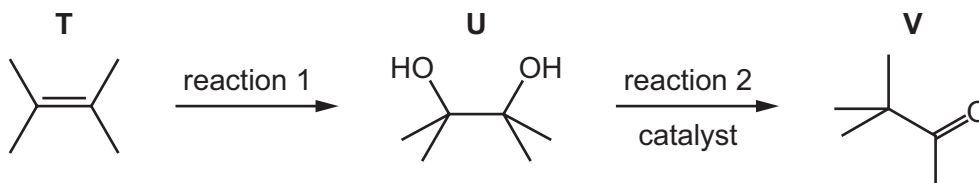
[2]

14. O/N 21/P22/Q3/e(vi,vii)

3 Compound **T** is an isomer of C_6H_{12} .

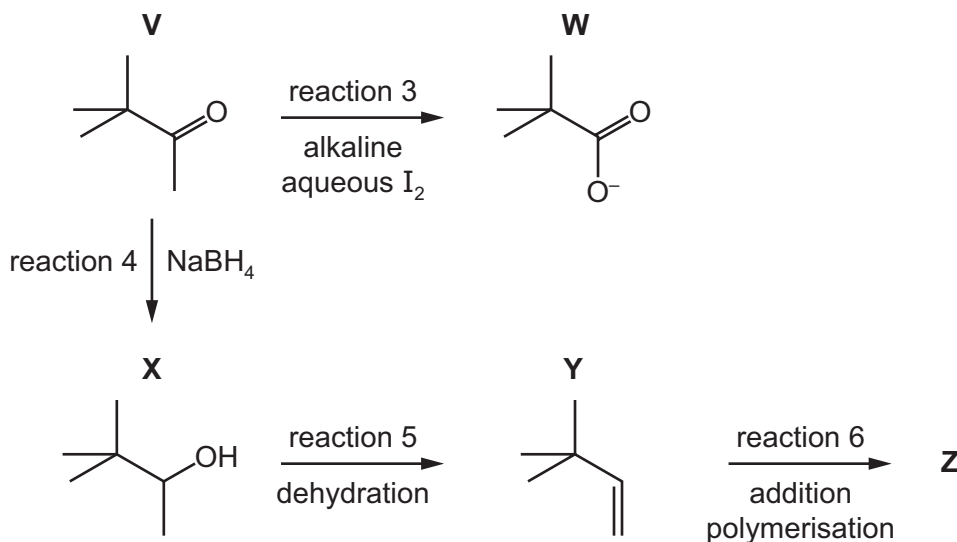


(d) A reaction scheme starting with **T** is shown. Reaction 2 occurs in the presence of a catalyst; knowledge of the mechanism for this reaction is not required.



(e) **V** is used in a wide range of organic reactions.

Some reactions of **V** are shown.



(vi) **Y** can form an addition polymer **Z**.

Draw one repeat unit of **Z**.

[1]

(vii) Reaction 6 does not proceed quickly at room temperature.

Suggest why this is the case.

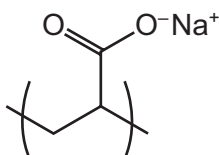
[1]

15. M/J 21/P21/Q5/d,e

- (c) Propene is separated from the mixture and heated in air in the presence of a catalyst. Propene is oxidised to **X**, which contains two functional groups.
- (d) **X** reacts with another reagent to form **Y**.

Molecules of **Y** react together to form addition polymer **Z**. The diagram shows the repeat unit of polymer **Z**.

repeat unit of polymer **Z**



Draw the structural formula of monomer **Y**.

[1]

- (e) Polymer **Z** is useful because it absorbs large amounts of water. However, there are problems associated with the disposal of products containing polymer **Z**.

Combustion is not an appropriate method to dispose of pure **Z** because the process releases harmful gases. Some of these gases contribute to the enhanced greenhouse effect.

- (i) Identify a gas released during the combustion of **Z** which contributes to the enhanced greenhouse effect.

[1]

- (ii) Identify another gas which could be produced during the combustion of pure **Z**. Describe a consequence, other than the enhanced greenhouse effect, of its release into the atmosphere.

gas

consequence

[1]

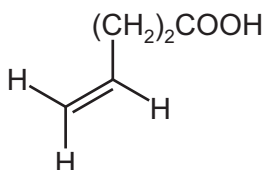
16. MAR 20/P22/Q3/c(ii)

- 3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

- (c) Fats are compounds made from glycerol and unsaturated carboxylic acids.

4-pentenoic acid is an example of an unsaturated carboxylic acid.

4-pentenoic acid

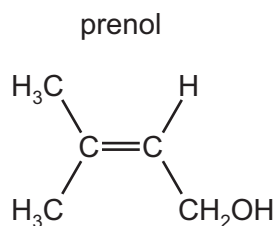


- (ii) Draw the repeat unit of the addition polymer that can be formed from 4-pentenoic acid.

[1]

17. O/N 19/P22/Q4/c(ii)

- 4 Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



- (ii) Prenol can be polymerised to form poly(prenol).

Draw **one** repeat unit of poly(prenol).

[1]

18. O/N 18/P21/Q3/c(i,ii,iv)

- 3 Trihalomethanes are organic molecules in which three of the hydrogen atoms of methane are replaced by halogen atoms, for example CHCl_3 .

- (c) CHClF_2 is also used to produce the monomer tetrafluoroethene, C_2F_4 .

This monomer can be used to produce poly(tetrafluoroethene), PTFE.

- (i) State the type of polymerisation that occurs during the production of PTFE.

[1]

- (ii) Draw the repeat unit of PTFE.

[1]

- (iv) Waste disposal can cause litter problems.

State two **other** difficulties associated with the disposal of PTFE.

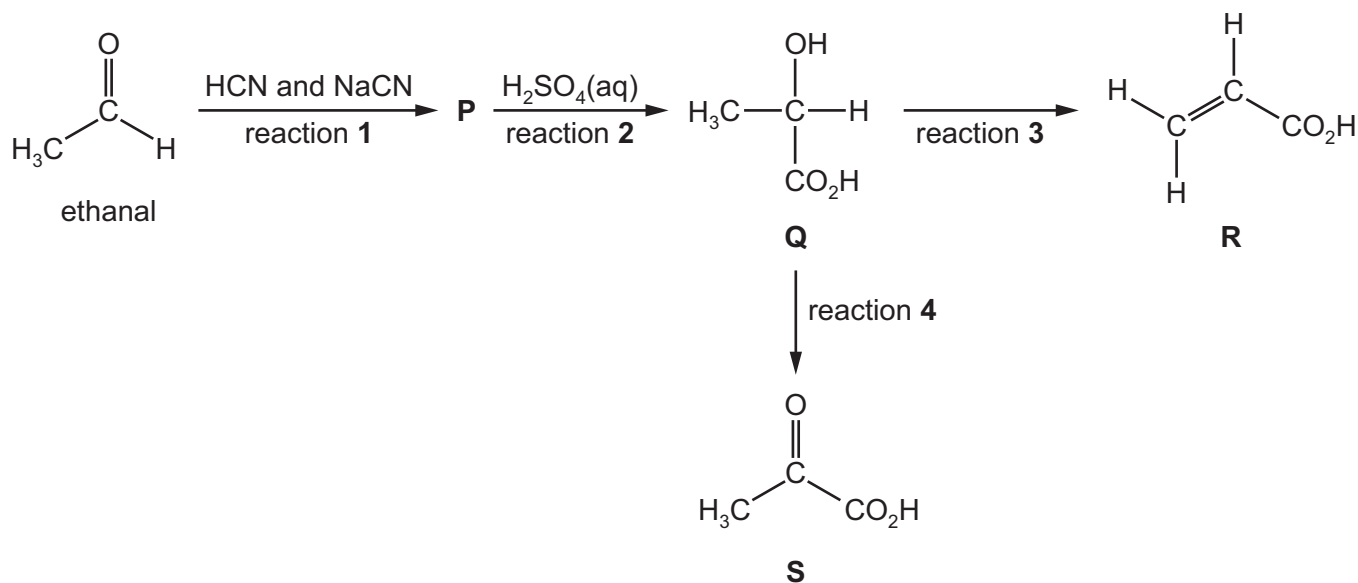
1

2

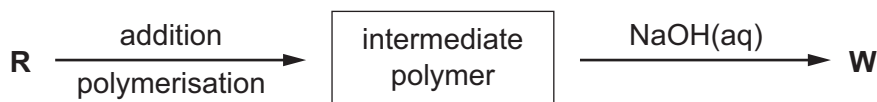
[2]

19. O/N 18/P22/Q4/c

4 The diagram shows a reaction sequence starting from ethanal.



(c) **R** can be used to make a polymer, **W**, in two steps.



Draw one repeat unit of **W**.

[3]

20. O/N 18/P23/Q3/c(i,ii,iv)

3 Trihalomethanes are organic molecules in which three of the hydrogen atoms of methane are replaced by halogen atoms, for example CHCl_3 .

(c) CHCl_2F_2 is also used to produce the monomer tetrafluoroethene, C_2F_4 .

This monomer can be used to produce poly(tetrafluoroethene), PTFE.

(i) State the type of polymerisation that occurs during the production of PTFE.

[1]

(ii) Draw the repeat unit of PTFE.

[1]

(iv) Waste disposal can cause litter problems.

State two **other** difficulties associated with the disposal of PTFE.

1

2

[2]

21. M/J 18/P21/Q2/c(ii,iii,iv)

- 2** Crude oil is a complex mixture of hydrocarbon molecules.

The hydrocarbon molecules in crude oil are separated by fractional distillation. Fractional distillation is used because the different hydrocarbon molecules in crude oil have different boiling points.

- (c)** Cracking one mole of dodecane, $C_{12}H_{26}$, produces two moles of ethene and one mole of another hydrocarbon molecule.

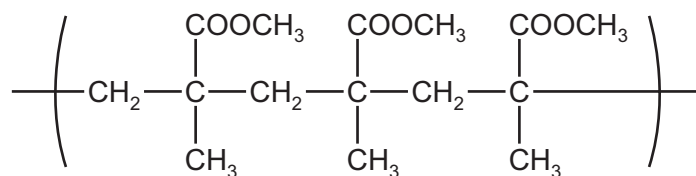
- (ii)** Give the full name of the process used to produce poly(ethene) from ethene.

[1]

- (iii)** Give **two** reasons why poly(ethene) should be reused or recycled rather than just thrown away.

[2]

- (iv)** Part of a polymer chain, produced by the same type of process as poly(ethene), is shown.



Give the **displayed** formula of the monomer used to produce this polymer.

[2]

22. M/J 18/P22/Q3/b(ii,iii)

- 3** Most vehicle fuels contain hydrocarbons obtained from crude oil.

- (b)** The supply of material suitable for use as fuels directly from crude oil is **not** sufficient to meet demand. A process is carried out to make some of the larger hydrocarbon molecules more useful.

- (ii)** Draw the repeat unit of the polymer that is produced from but-2-ene.

[2]

- (iii)** Name the type of polymerisation that occurs during the production of the polymer in **(ii)**.

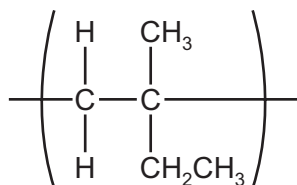
[1]

23. M/J 18/P23/Q2/b(ii)

- 2** One reason for the wide variety of organic compounds is isomerism, either structural isomerism or stereoisomerism.

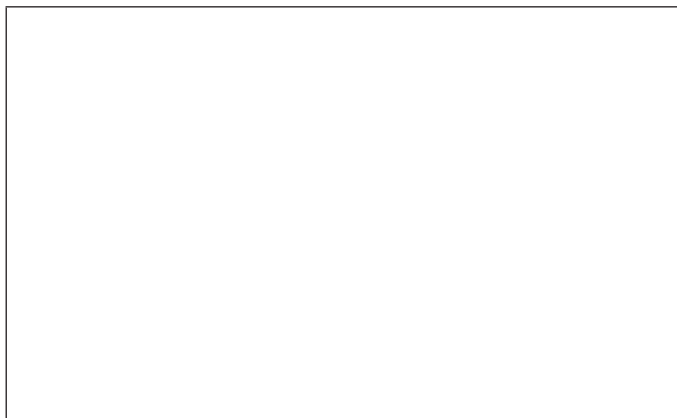
(b) Pent-1-ene, $\text{CH}_2=\text{CH}(\text{CH}_2)_2\text{CH}_3$, does not show stereoisomerism.

- (ii)** A structural isomer of pent-1-ene is used as the monomer to form a polymer. The repeat unit of this polymer is shown.



Draw the **displayed** formula of the monomer used to make this polymer.

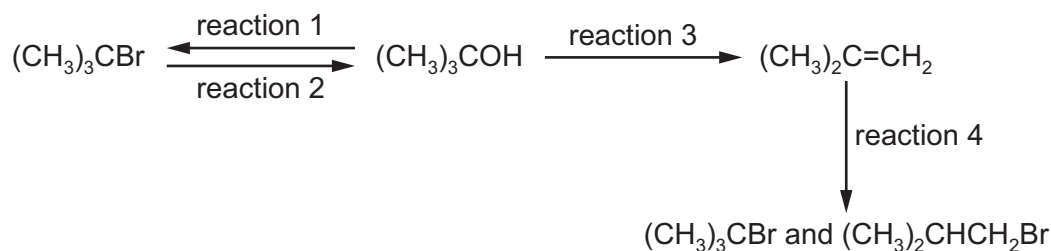
Give the name of the monomer.



[2]

24. O/N 17/P22/Q4/d(ii,iii)

4 Some reactions are shown, based on methylpropan-2-ol, $(\text{CH}_3)_3\text{COH}$.



(d) The product of reaction 3, methylpropene, does **not** show stereoisomerism.

(ii) Methylpropene can be polymerised to form a poly(alkene).

State the type of polymerisation and draw the repeat unit of the polymer formed from methylpropene.

type of polymerisation

repeat unit

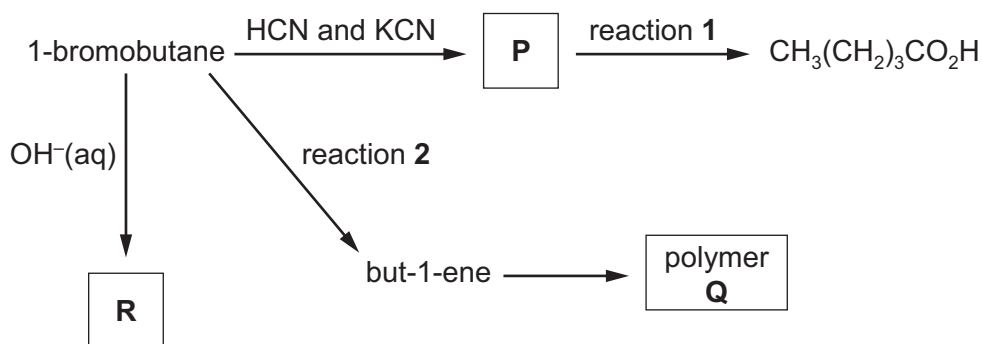
[3]

(iii) State the difficulty associated with the disposal of poly(alkenes).

[1]

25. MAR 17/P22/Q3/a(iii)

3 (a) A series of reactions starting from 1-bromobutane is shown.



(iii) Draw the structure of the repeat unit of polymer Q.

[2]

26. O/N 16/P21/Q4/b(iv)

4 In each section of this question an organic compound is shown. For each compound give its name and answer the questions about it.

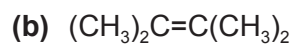
(b) $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)_2$

(iv) Draw the structure of part of a molecule of the addition polymer formed from this compound, showing exactly **three** repeat units.

[1]

27. O/N 16/P23/Q4/b(iv)

- 4** In each section of this question an organic compound is shown. For each compound give its name and answer the questions about it.

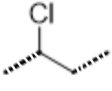
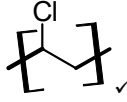


- (iv)** Draw the structure of part of a molecule of the addition polymer formed from this compound, showing exactly **three** repeat units.

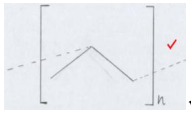
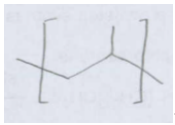
[1]

Answers Section

1. M/J 26/P21/Q5/f

5(f)	$-\text{[CH}_2\text{-CHCl]}\text{-}$ OR <pre> H Cl - C - C - H H </pre> OR <pre> H H - C - C - Cl H </pre>	1 I brackets (any) I n A Cl on either C atom A dangling bonds / bracketed repeat unit If skeletal then dangling bonds are different (e.g. dashed, v. long) or go through brackets  ✓ OR  ✓
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2. M/J 26/P24/Q5/b(i,ii)

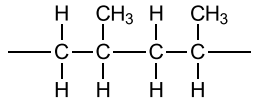
5(b)(i)	<pre> CH3 H (C - C) H H n </pre>	1 I brackets I n A skeletal structure <u>with</u> hanging bonds / identifiable continuation bonds (i.e. dashed lines):  ✓  ✓
5(b)(ii)	non-biodegradable	1 I produces harmful products / toxic gas I other environmental effects I disposal / cost A non-photodegradable A unreactive due to strong / non-polar bonds A phonetical spelling A BOD difficult / hard to decompose / does not decompose / does not break up / down (easily)

3. O/N 25/P21/Q4/b(i,iii,iv)

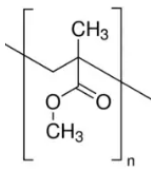
4(b)(i)	addition	1
4(b)(iii)	<pre> H CH3 H CH3 - C - C - C - C - H H H H </pre> M1 M2 optical isomerism M3 the methylated carbon has four different groups attached	3
4(b)(iv)	M1 non-biodegradability M2 harmful combustion products	2

4. O/N 25/P23/Q4/b(i,iii,iv)

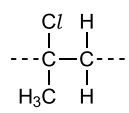
4(b)(i)	addition	1
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4(b)(iii)	 M1 M2 optical isomerism M3 the methylated carbon has four different groups attached	3
4(b)(iv)	M1 non-biodegradability M2 harmful combustion products	2

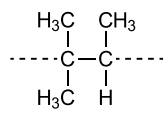
5. M/J 25/P23/Q5/b(vi),c

5(b)(vi)		1
5(c)	ethene pi bond is a region of (high) electron density (above and below the C-C (sigma) bond)	1

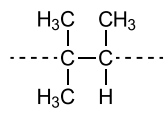
6. MAR 24/P22/Q2/d(iii)

2(d)(iii)		1

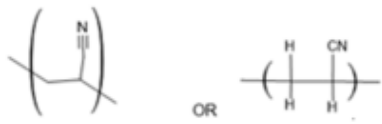
7. O/N 23/P21/Q4/a(v)

4(a)(v)		1
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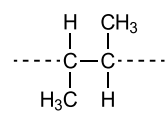
8. O/N 23/P23/Q4/a(v)

4(a)(v)		1
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9. M/J 23/P23/Q6/a

6(a)		1
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10. MAR 23/P22/Q4/a(ii)

4(a)(ii)		1
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11. O/N 22/P22/Q3/b

3(b)(i)	addition	1
3(b)(ii)	$\begin{array}{c} \text{H} \quad \text{CH}_2\text{CH}_2\text{OH} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{H} \end{array}$ M1 correct carbon backbone of one repeat unit including 'dangling' bonds	1
	M2 rest of structure correct	1

12. M/J 22/P22/Q4/c

4(c)(i)		1
4(c)(ii)	non-biodegradable OR poisonous / harmful HCl made when burnt	1

13. MAR 22/P22/Q4/b(ii)

4(b)(ii)	$\begin{array}{c} \text{H} \quad \text{COOCH}_3 \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{CH}_3 \end{array}$ carbon backbone with 'dangling' bonds [1] rest of structure correct [1]	2
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14. O/N 21/P22/Q3/e(vi,vii)

3(e)(vi)	$\begin{array}{c} \text{H} \quad \text{C}(\text{CH}_3)_3 \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{H} \end{array} \quad \text{OR} \quad \begin{array}{c} \text{H} \quad \text{C}(\text{CH}_3)_3 \\ \cdots - \text{C} - \text{C} - \cdots \\ \quad \\ \text{H} \quad \text{H} \end{array}$	1
3(e)(vii)	high activation energy	1

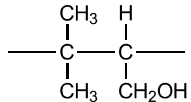
15. M/J 21/P21/Q5/d,e

5(d)	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}=\text{C}-\text{COONa} \\ \\ \text{H} \end{array}$ OR $\text{CH}_2\text{CH}(\text{CO}_2^{(-)}\text{Na}^{(+)})$ OR $\text{CH}_2=\text{CH}(\text{COO}^{(-)}\text{Na}^{(+)})$ $\begin{array}{c} \text{O} \\ \\ \text{O}=\text{C}-\text{O}^-\text{Na}^+ \end{array}$ OR $\begin{array}{c} \text{O}=\text{C}-\text{O}^-\text{Na}^+ \\ \\ \text{H}-\text{C}=\text{C}-\text{H} \\ \\ \text{H} \end{array}, \text{ etc.}$	1
5(e)(i)	carbon dioxide / CO_2	1
5(e)(ii)	CO / hydrocarbons AND toxic / poisonous / harmful to health / (catalyses formation of) photochemical smog	1

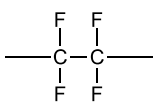
16. MAR 20/P22/Q3/c(ii)

3(c)(ii)	$\begin{array}{c} \text{H} \quad (\text{CH}_2)_2\text{COOH} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \quad \text{H} \end{array}$	1
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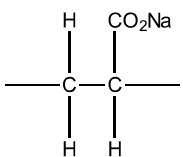
17. O/N 19/P22/Q4/c(ii)

4(c)(ii)		1
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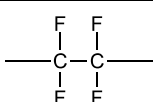
18. O/N 18/P21/Q3/c(i,ii,iv)

3(c)(i)	addition	1
3(c)(ii)		1
3(c)(iv)	non-biodegradable creates toxic / harmful gases / HF / CO ₂ / CO if burnt	2

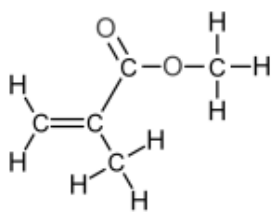
19. O/N 18/P22/Q4/c

4(c)	 OR -CH ₂ CH(CO ₂ Na)-	3
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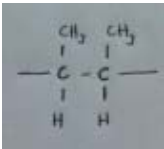
20. O/N 18/P23/Q3/c(i,ii,iv)

3(c)(i)	addition	1
3(c)(ii)		1
3(c)(iv)	non-biodegradable creates toxic / harmful gases / HF / CO ₂ / CO if burnt	2

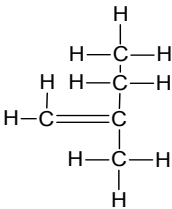
21. M/J 18/P21/Q2/c(ii,iii,iv)

2(c)(ii)	addition polymerisation	1
2(c)(iii)	<i>two from</i> save space in landfill avoid litter prevent eyesore non-biodegradable conserves non-renewable resources harmful incineration products harmful to wildlife	2
2(c)(iv)		
	correct monomer	1
	fully displayed	1

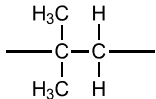
22. M/J 18/P22/Q3/b(ii,iii)

3(b)(ii)	$\text{-CH(CH}_3\text{)CH(CH}_3\text{)-}$ OR 	
	even number of C's in correct backbone with 'end bonds' AND no C=C	1
	CH ₃ and H on each of two C drawn for 1 repeat unit only AND all the carbons must be tetravalent	1
3(b)(iii)	addition	1

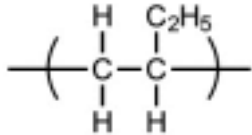
23. M/J 18/P23/Q2/b(ii)

2(b)(ii)		1
	2-methylbut-1-ene	1

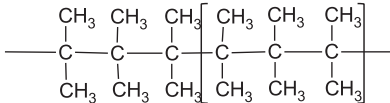
24. O/N 17/P22/Q4/d(ii,iii)

4(d)(ii)	addition	1
	 marking points: - correct number of tetravalent carbon atoms in backbone, with extension bonds - correct groups on backbone carbon atoms and only one repeat unit	2
4(d)(iii)	not/non- biodegradable / harmful combustion products	1

25. MAR 17/P22/Q3/a(iii)

3(a)(iii)		2
	C-C backbone with dangling bonds rest of structure	1 1

26. O/N 16/P21/Q4/b(iv)

4(b)(iv)		1	1
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27. O/N 16/P23/Q4/b(iv)

4(b)(iv)	$\begin{array}{ccccccc} & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 \\ & & & & & & \\ - & \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{C} - \\ & & & & & & \\ & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 & \text{CH}_3 \end{array}$	1	1
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